Arjun Nair

AI/ML & Full-Stack Engineer

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PROFILE

Second-year AI/ML undergraduate with four internships at EY, Sypha AI and Trezix, building complete products across LLM pipelines, real-time data systems and native apps.

EXPERIENCE

Sypha AI Software Development Intern — Oct 2025 – Dec 2025
- Built a Flask dashboard using GPT-4o-mini as a judge to score prompts and rank models by accuracy.

EY Generative AI Intern — May 2025 – Aug 2025
- Built LangChain agents that turn equipment manuals into structured maintenance knowledge bases.
- Delivered RAG chatbots over those knowledge bases using embeddings with FAISS and ChromaDB vector stores.

Trezix AI/ML Intern — Mar 2025
- Applied ML models to banking datasets to produce profit-optimisation strategies.

EY Data Analyst Intern — Jun 2023 – Jul 2023
- Built a Python translation tool and shipped it as a web app supporting more than 50 languages.

PROJECTS

Pit Wall Live F1 timing and race-strategy platform — GitHub
Next.js, TypeScript, FastAPI, WebSockets, PostgreSQL, Redis, ChromaDB, LangGraph, Docker
- Streams F1's live timing feed into a dashboard that forecasts DRS battles and pit-stop rejoin positions; on a replay of Singapore 2026, 21 of 23 real stops landed within one place of the forecast.
- Strategy simulator, undercut finder, lap-by-lap win probability and Monte Carlo title odds since 2018. LLMs only word numbers they are given, and each sentence is checked against the data. 200+ commits with full CI.

PurpleSector Sim-racing telemetry and voice race engineer — GitHub
Python, UDP binary protocols, Streamlit, SQLite, Piper neural TTS, PyInstaller
- One decoder for four years of F1 game telemetry formats, tested against each spec, logging 60 Hz data for 22 cars.
- Compares laps metre by metre and radios time lost per corner before the next one; shipped as a Windows app.

Jewlerify Verified-jewellery marketplace — Live site GitHub
React 19, TypeScript, Tailwind, FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT, Vercel
- Each piece carries a QR tag; scanning shows its purity, weight and a price itemised from the live gold rate.
- Exact decimal pricing with database constraints, photo metadata stripping and on-device buyer location.

APEX Native F1 widgets for iOS, Android and Windows — GitHub
Swift / WidgetKit, Kotlin, C# / WinUI, FastAPI, OpenF1
- Ten widgets with Live Activities and Dynamic Island that switch from countdown to live timing to results.

Resonance — GitHub
Next.js, FastAPI, PostgreSQL, Redis, Docker
Music platform with a hybrid recommender blending content, collaborative, behaviour and mood signals.

FitRank Pro — Live demo GitHub
Java, PostgreSQL, JavaScript
Workout tracker with PR detection and 1RM trends; imports Apple Health and Hevy history.

CalTrack — GitHub
TensorFlow / Keras, OpenCV, Node.js, MongoDB
Calorie tracker with a CNN food classifier trained from scratch, reaching 84.5% validation accuracy.

TradeBot — GitHub
Python, REST, Click
Binance Futures trading CLI with hand-written HMAC-SHA256 request signing and pre-flight order validation.

Microfinance Loan System — GitHub
Java, JDBC, MySQL
Loan management with EMI scheduling, repayment tracking and credit evaluation on a layered DAO design.

Resume Parser — GitHub
Python, python-docx
Converts PDF and DOCX resumes into structured JSON for automated scoring and recommendations.

SKILLS

Languages	Python, TypeScript, JavaScript, Java, SQL, Swift, Kotlin, C#
AI / ML	LangChain, LangGraph, RAG, OpenAI and Claude APIs, TensorFlow, pandas, NumPy
Engineering	FastAPI, Flask, Next.js, React, Tailwind, PostgreSQL, Redis, MongoDB, MySQL, WebSockets
Tooling	Docker, GitHub Actions, Vercel, Azure, Jenkins, Git

EDUCATION

Atlas Skilltech University B.Tech, Artificial Intelligence and Machine Learning — 2024 – 2028

IIT Mandi Minor in AI and Data Science, 15 credits — 2025 – 2026

Singapore International School International Baccalaureate Diploma — 2022 – 2024

HONOURS

Research paper on bias mitigation in ML selected for Springer publication (2025). Subject topper in Model Thinking, Statistical Modelling, Databases and SQL, and Computer Organisation. Co-Head, SPARC Esports; Kaleidoscope Premier League winner; Inspirit AI Scholar.